



Sri Lanka Institute of Information Technology

A Modern Digital Storefront for Gen-Z

Sprint 0

Information Systems Project Management 2026

Project ID: ISE_WE_0201_58

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Date of submission

25/08/2026

Criteria	Marks allocated	Marks given
Problem Specification	5%	
Business Idea	5%	
Solution Outline	10%	
Key Benefits	10%	
Main Objectives	5%	
Sub Objectives	5%	
Project Plan	10%	
Product Vision statement	05%	
Epics	05%	
User Stories	10%	
Initial product Backlog	05%	
Definition of Done	05%	
Hardware & Software requirements	5%	
Budget	5%	
References	5%	
Formatting	5%	
Total Marks (100)	100%	

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1. Introduction

1.1 Problem Specification

Gen-Z is a men's, boys' clothing and perfume retail store that currently operates primarily through a physical outlet, relying on in-person visits and informal, ad-hoc communication telephone calls and word of mouth to manage customer enquiries, orders and product information. This dependence on manual, offline processes restrict the store's reach to customers who cannot visit in person, creates inconsistent and error-prone product and order records, and gives the store owner limited real-time visibility into sales, stock and customer feedback.

Without a centralized product catalogue, product details, pricing and stock levels are tracked manually, increasing the risk of errors, outdated listings and lost sales from items shown as available when they are not. Orders placed informally over calls or messages are not consistently recorded, making it difficult to track order status, coordinate delivery handovers, monitor delivery progress, or analyse sales trends. There is currently no structured way for customers to leave reviews or ratings, limiting the store's ability to build online trust or gather feedback for product improvement.

The owner has no single dashboard to monitor products, orders, reviews and overall store activity, forcing reliance on memory and manual cross checking. These issues directly limit revenue growth, increase the owner's operational workload, and put Gen-Z at a competitive disadvantage relative to fashion retailers that already offer a structured online presence. A purpose-built, web-based e commerce platform is required to digitise and streamline the store's core processes across product management, order handling, delivery tracking, customer reviews and administrative oversight.

1.2 Business idea

This project proposes the development of a modern, responsive web-based e-commerce platform tailored specifically for Gen-Z a men's, boys' clothing and perfume retail store operating in the Sri Lankan market. The platform will provide customers with a seamless digital experience for browsing a curated product catalogue, viewing detailed product and pricing information, and placing orders through a familiar WhatsApp-based checkout workflow. A delivery tracking mechanism will give customers real-time visibility into their order status from confirmation through to delivery, while a structured review and rating system will help build trust and drive repeat purchases.

The core business opportunity lies in transforming an informal, geography-limited retail operation into a digitally accessible storefront available to customers at any time and from any device. By deliberately excluding an online payment gateway and a native mobile application in Phase 1, the system is designed to be practically deployable within the constraints of a small retail operation while still delivering immediate and measurable business value. The platform equips the store owner with a self-service administration dashboard to manage products, monitor inventory, process and track orders, assign delivery personnel, moderate customer reviews and configure store setting all without requiring technical expertise.

1.3 Solution Outline

The proposed solution is a full-stack, web-based e-commerce application developed using a modern three-tier architecture. The system is structured into three primary layers: a React.js frontend for the customer-facing storefront and administrator dashboard, a Node.js and Express.js RESTful API backend for business logic and data processing, and a MySQL relational database for persistent data storage. Product images are stored and served via Cloudflare R2 object storage, providing a scalable and cost-effective media management solution.

The system comprises two distinct user-facing interfaces: a public storefront accessible to all customers without mandatory registration for browsing, and a protected administrator dashboard accessible exclusively to authorized store personnel. The checkout process follows a WhatsApp-based workflow, wherein the customer's cart details are automatically compiled into a pre-formatted message and routed to the store owner's WhatsApp number. Once an order is confirmed and prepared, it is assigned to a delivery person whose status update Picked Up, Out for Delivery, Delivered are tracked through the platform, giving customers transparent visibility into their delivery progress.

The high-level system architecture is illustrated in the diagram below and organized into five logical layers:

- Client Layer: Customer storefront (product browsing, search, cart, WhatsApp checkout, order and delivery tracking, reviews) and Admin dashboard (product management, order management, delivery assignment, review moderation, user management, store configuration).

- API Gateway & Backend: RESTful endpoints for all business logic, JWT-based authentication and role-based access control, image handling, and WhatsApp message generation.
- Application Server Business Logic Modules: EP-01 Product & Catalogue Management; EP-02 Customer & Order Management; EP-03 Delivery Tracking with Review Management; EP-04 Store Administration Management.
- Database Layer: MySQL relational schema storing products, categories, orders, delivery assignments, customer accounts, reviews, moderation logs, and administrative configurations.
- External Services: Cloudflare R2 for product image storage and delivery; WhatsApp for the customer checkout communication flow.

Business Architecture Overview

The diagram below provides a complete business architecture view of the Gen-Z platform, illustrating the four core business capability modules (EP-01 through EP-04), the key actors (Customer, Delivery Person, Store Owner / Admin), the data flows between modules, and the external services (WhatsApp checkout channel and Cloudflare R2 image storage) that underpin the platform's end-to-end operation.

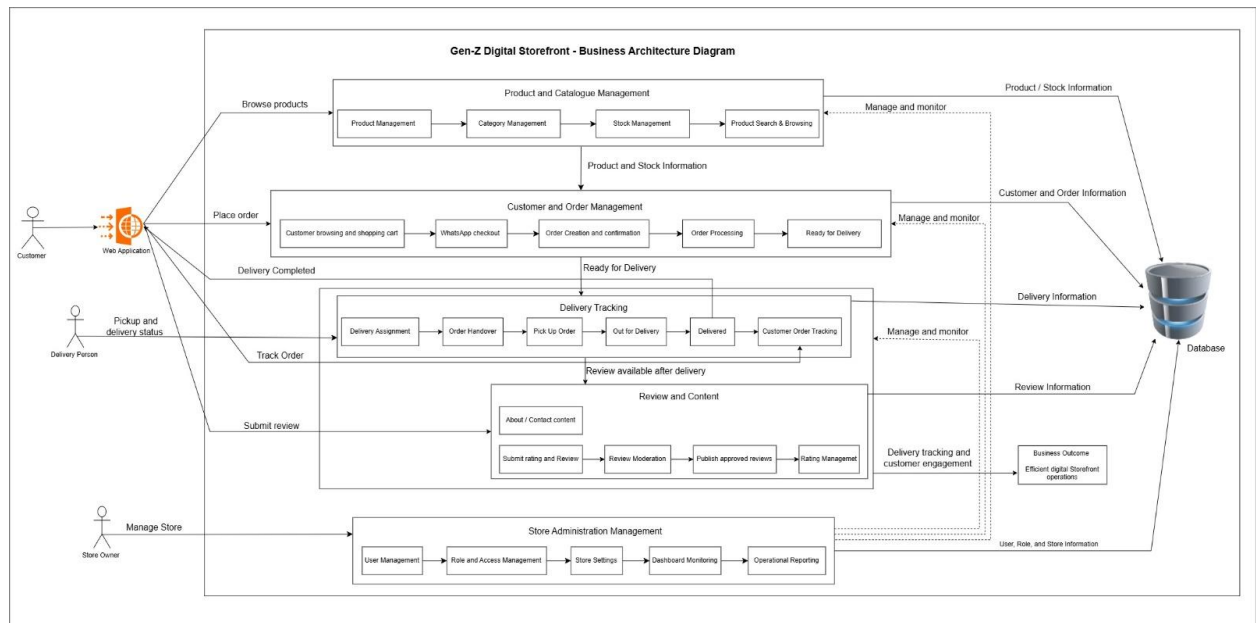


Figure 1: Business Architecture Diagram – A Modern Digital Storefront for Gen-Z

1.4 Key Benefits

- **24/7 Digital Accessibility:** Customers can browse men's clothing, boys' clothing and perfumes, check stock availability, and initiate orders at any time from any internet-connected device, removing the constraint of physical store operating hours.
- **Streamlined Order Processing:** The WhatsApp-based checkout flow eliminates unstructured verbal or ad hoc messaging orders, ensuring that complete and accurate order information is captured and stored with every transaction.
- **End-to-End Delivery Visibility:** The integrated delivery tracking workflow — from order assignment through Picked Up, Out for Delivery, to Delivered — gives customers transparent, real-time visibility into their delivery status, reducing enquiries to the store.
- **Reduced Manual Data Entry and Errors:** Centralised product and order management eliminates repeated manual data entry across multiple channels, reducing the risk of stock discrepancies, duplicate listings and order fulfillment errors.
- **Improved Customer Trust and Engagement:** A verified-purchase review and rating system with admin moderation enables customers to make more informed purchase decisions, increasing purchase confidence and brand credibility.
- **Operational Visibility for Store Management:** The admin dashboard provides the store owner with a unified view of all products, orders, delivery statuses, reviews and customer activity, supporting data-driven decisions for stock replenishment and business planning.
- **Role-Based Access Control:** Staff and admin accounts with defined permissions ensure that only authorised personnel can perform sensitive operations, reducing the risk of unauthorised changes or data exposure.
- **Scalable Platform Architecture:** The three-tier architecture with cloud-based image storage is designed to accommodate future enhancements including online payment integration, expanded product categories and multi-branch support, without requiring a full system redesign.

2. Objectives

2.1 Main Objective

To design and develop a functional, web-based e-commerce platform for Gen-Z a men's, boys' clothing and perfume retail store that digitises the store's product catalogue management, WhatsApp-based customer ordering, delivery tracking, customer review management and administrative operations, enabling the store owner to reach a broader customer base, provide end-to-end order and delivery visibility, and manage all store activities through a centralised self-service dashboard, within a thirteen-week academic project timeline.

2.2. Sub Objectives

- To implement a fully functional product and catalogue management module (EP-01) that allows the store administrator to create, update, categorise and delete listings for men's clothing, boys' clothing and perfumes, with associated images, verified by successful admin dashboard functional testing within the first sprint.
- To develop a customer-facing shopping cart and WhatsApp-based checkout mechanism (EP-02) that accurately compiles and routes order details to the store owner, with order records persisted and status tracked through Pending, Confirmed, Processing and Ready for Delivery, validated through end-to-end integration testing before the Sprint 2 review.
- To build a delivery tracking and review management module (EP-03) that enables delivery person assignment, status tracking (Assigned, Picked Up, Out for Delivery, Delivered), customer-visible order tracking and admin-moderated review submission and publishing, verified by functional and integration testing in Sprint 3.
- To deliver a store administration management module (EP-04) with role-based access control ensuring only authenticated and authorised users can perform administrative operations, a unified dashboard summarising all operational areas, and store-wide settings configuration, confirmed through security and access control testing in the final sprint.
- To deploy the completed application to a live, publicly accessible URL with an accompanying user guide and technical documentation delivered by the end of the thirteen-week project period.

3. Stakeholder Identification

List and categorize key stakeholders:

Stakeholder	Type	Interest / Concern
Store Owner / Administrator	Primary	Full control over products, orders, delivery management, review moderation and store configuration; operational efficiency and revenue growth.
Customers	Primary	Easy product discovery across clothing and perfumes, accurate stock information, familiar WhatsApp checkout, transparent delivery tracking, and peer-reviewed trust signals.
Delivery Person	Primary	Clear order assignment, delivery details, and a straightforward mechanism for updating delivery status through the platform.
Sales / Floor Staff	Secondary	Clear guidance on new order and dispatch workflows; access to relevant order information within assigned permissions.
Project Development Team	Primary	Successful design, development, testing and delivery of the platform within the academic timeline and quality standards.
Academic Supervisor / Lecturer	Secondary	Project adherence to academic requirements, methodology, documentation quality and assessment criteria.
Suppliers	Tertiary	Continued stock supply to the store; no direct system interaction in Phase 1.
Cloudflare R2 / Hosting Provider	Tertiary	Reliable infrastructure availability; SLA compliance for image storage and hosting services.
WhatsApp / Meta	Tertiary	Platform availability and messaging protocol stability for the WhatsApp-based order checkout integration.

4. Procedure

Development Methodology

This project follows the Scrum agile framework, which was selected for three key reasons. First, the project scope is subject to refinement of Scrum's iterative sprint structure allows requirements to be clarified and adjusted after each review without disrupting the overall delivery plan. Second, the three-member team size is well suited to Scrum's lightweight role structure (Product Owner, Scrum Master, Team Member). Third, Scrum's sprint-based delivery model ensures that a working, demonstrable increment of the platform is produced every two to three weeks, reducing the risk of a late-stage integration failure.

Assumptions

- The store owner (Gen-Z) has a functioning WhatsApp account and a stable mobile internet connection for receiving and responding to customer orders.
- Customers have access to WhatsApp on their mobile devices and are familiar with its use for communication.
- Cloudflare R2 and the selected cloud hosting platforms remain available and within free-tier limits throughout the thirteen-week project period.
- The development team has uninterrupted access to their development workstations and the internet throughout the project timeline.
- All three team members are available for the full thirteen-week duration with no prolonged absences.

Constraints

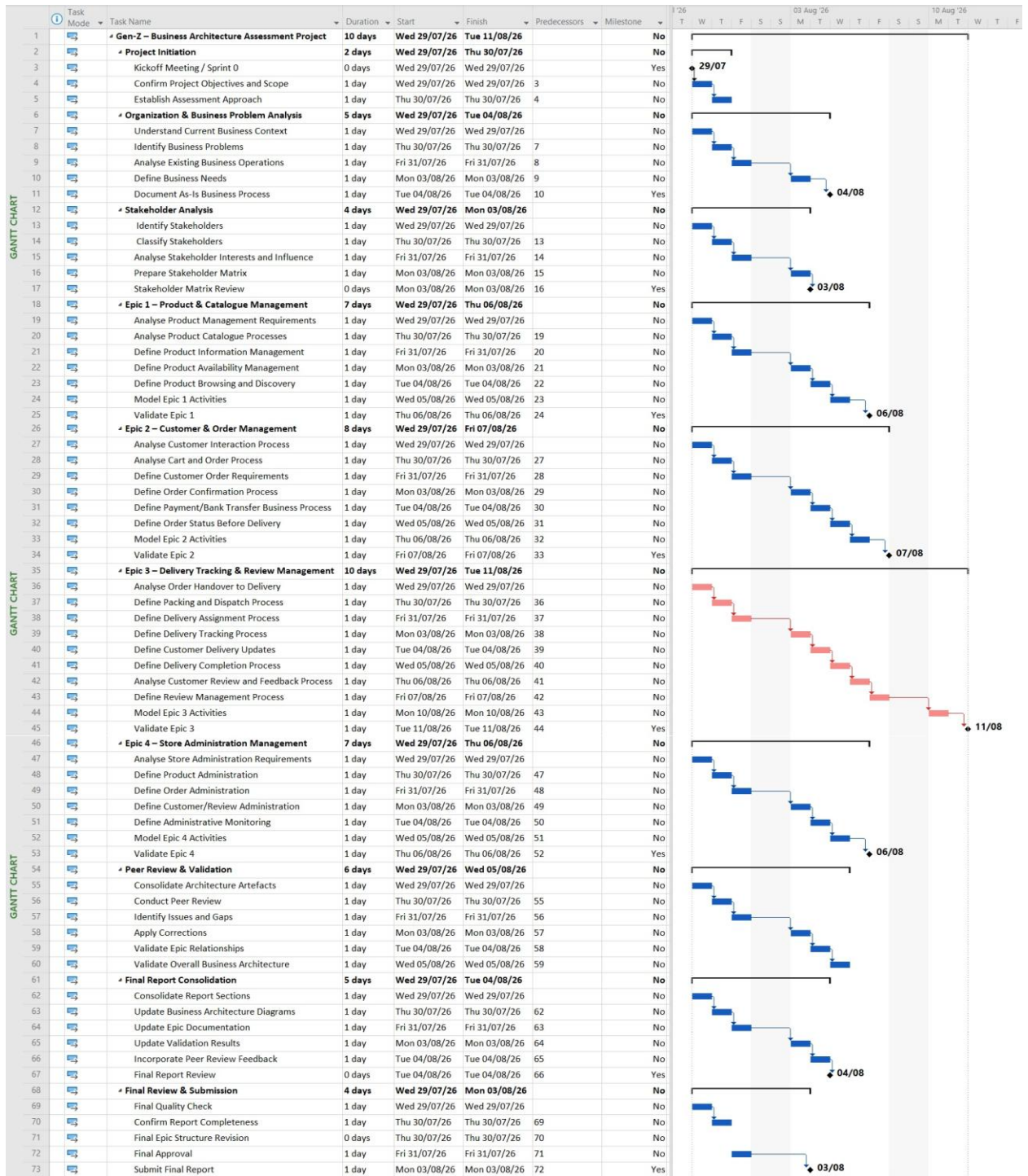
- **Time:** The project will be completed within 13 weeks, covering five sprints.
- **Scope:** Phase 1 will not include an online payment gateway, native mobile application, delivery logistics management beyond basic status updates, or a refund and dispute resolution system.
- **Technology:** The system will be developed as a web-based platform and accessed through desktop and mobile browsers.
- **Team:** The development team has three members, so development activities will need to be planned according to the available team capacity.
- **Budget:** The project will use free-tier or low-cost cloud services wherever possible. The estimated total project cost is LKR 7,450.

Sprint Schedule

The following table summarizes the planned sprint schedule and key deliverables for each sprint:

Sprint	Key Deliverables / Focus Areas	Duration
Sprint 0	Project Charter, proposal, product backlog, epics, user stories, Definition of Done	Week 1–2
Sprint 1	Project setup, DB schema design, EP-01: Product & Catalogue CRUD, EP-04: Admin user accounts & RBAC (foundation)	Week 3–5
Sprint 2	EP-02: Customer storefront, shopping cart, WhatsApp checkout, order record creation, status Pending → Ready for Delivery	Week 6–8
Sprint 3	EP-03: Delivery assignment & tracking (Assigned → Picked Up → Out for Delivery → Delivered), review submission & moderation	Week 9–10
Sprint 4	EP-04: Admin dashboard, RBAC completion, store settings, activity logs, final integration testing, deployment, documentation	Week 11–13

The Gantt chart below illustrates the sprint schedule, key tasks and milestone dates across the thirteen-week project timeline:



The product A Modern Digital Storefront for Gen-Z is a responsive, web-based e-commerce platform,

That enables customers to discover, evaluate and order products online through a WhatsApp-based checkout with end-to-end delivery tracking, while providing the store owner with a centralized self-service dashboard to manage the full product catalogue, customer orders, delivery assignments, review moderation, and store operations efficiently and independent

6. Epics

Epic ID	Epic Description
EP-01	Product & Catalogue Management Manages the complete product catalogue for men's clothing, boys' clothing and perfumes, including creation, categorisation, image management and customer-facing product discovery. Supports the store owner in maintaining accurate product information and provides customers with a seamless browsing and search experience.
EP-02	Customer & Order Management Manages the end-to-end customer ordering process from cart creation through WhatsApp-based checkout to admin-side order tracking, ending when an order is marked Ready for Delivery. Ensures accurate order capture and full store owner visibility of all orders and statuses.
EP-03	Delivery Tracking with Review Management Manages delivery person assignment, delivery status tracking (Assigned, Picked Up, Out for Delivery, Delivered), customer-visible order tracking, and customer-generated product reviews with admin moderation controls. Builds customer trust and completes the order fulfilment lifecycle.
EP-04	Store Administration Management Manages centralised administrative control over the platform including staff user account management, role-based access controls, a unified admin dashboard summarising all operational areas, store-wide settings configuration, and activity logging.

7. User Stories

Epic ID	Story ID	User Story	Stakeholder
EP-01	US-01	As a store administrator, I want to add new products (men's clothing, boys' clothing and perfumes) with images, descriptions, price and stock quantity, so that customers can discover and purchase the latest items in the store.	Store Owner
EP-01	US-02	As a store administrator, I want to edit existing product details including price, description and stock quantity, so that the product catalogue remains accurate and up to date.	Store Owner

EP-01	US-03	As a store administrator, I want to delete products from the catalogue, so that discontinued or unavailable items are no longer displayed to customers.	Store Owner
EP-01	US-04	As a store administrator, I want to organise products into categories (e.g., men's clothing, boys' clothing, perfumes) and apply tags, so that customers can navigate and filter products more easily.	Store Owner
EP-01	US-05	As a customer, I want to browse the product catalogue with filtering by category, price range and name, so that I can quickly find products that match my preferences.	Customer
EP-01	US-06	As a customer, I want to view a detailed product page with images, description, price and stock availability, so that I can make an informed purchase decision.	Customer
EP-01	US-07	As a store administrator, I want the system to automatically reflect In Stock and Out of Stock status based on inventory levels, so that customers cannot order unavailable products.	Store Owner / Customer
EP-02	US-08	As a customer, I want to add products to a shopping cart and adjust quantities before checkout, so that I can manage my intended purchase before confirming.	Customer
EP-02	US-09	As a customer, I want the system to generate a pre-filled WhatsApp message containing my full order details, so that I can send it directly to the store owner to confirm my order without re-entering information.	Customer
EP-02	US-10	As a customer, I want my order to be automatically recorded in the system when I initiate the WhatsApp checkout, so that the store owner can track and manage my order from the dashboard.	Customer / Store Owner
EP-02	US-11	As a store administrator, I want to view all customer orders with their current statuses from the admin dashboard, so that I have full visibility of all pending and active orders.	Store Owner
EP-02	US-12	As a store administrator, I want to update the order status through the workflow (Pending → Confirmed → Processing → Ready for Delivery), so that internal records reflect the current fulfilment progress.	Store Owner
EP-02	US-13	As a customer, I want to view my order history and the current status of each order, so that I am kept informed	Customer

		about the progress of my purchases up to the point of dispatch.	
EP-03	US-14	As a store administrator, I want to assign a delivery person to a packed order, so that delivery responsibility is clearly recorded and the order moves into the delivery tracking workflow.	Store Owner
EP-03	US-15	As a delivery person, I want to accept an assigned order and update its status as I collect and deliver it (Assigned → Picked Up → Out for Delivery → Delivered), so that the store and customer have accurate real-time delivery visibility.	Delivery Person
EP-03	US-16	As a customer, I want to track the real-time delivery status of my order (Assigned, Picked Up, Out for Delivery, Delivered), so that I know when to expect my package without needing to contact the store.	Customer
EP-03	US-17	As a registered customer, I want to submit a star rating and written review for a product I have received, so that I can share my experience and help other customers make informed decisions.	Customer
EP-03	US-18	As a customer, I want to view approved reviews and aggregate star ratings on product pages, so that I can evaluate product quality before purchasing.	Customer
EP-03	US-19	As a store administrator, I want to approve, reject or delete customer-submitted reviews from a moderation panel, so that only appropriate and verified content is displayed publicly on the platform.	Store Owner
EP-03	US-20	As a store administrator, I want to manage the content of static pages such as the About and Contact sections, so that store information remains current and accurate for customers.	Store Owner
EP-04	US-21	As a store owner, I want to create, view, update, activate and deactivate system user accounts for admin and staff roles, so that access to the platform is properly managed and controlled.	Store Owner
EP-04	US-22	As a store owner, I want to assign roles and access permissions to system users, so that only authorised personnel can perform specific administrative operations.	Store Owner
EP-04	US-23	As a store owner, I want a unified admin dashboard summarising key metrics including product counts,	Store Owner

		pending orders, delivery statuses and pending reviews, so that I can quickly assess the operational status of the store.	
EP-04	US-24	As a store administrator, I want to configure store-wide settings including the store name, contact information and WhatsApp number, so that customer-facing information remains consistent and accurate.	Store Owner
EP-04	US-25	As a store owner, I want to view activity logs for key system operations, so that I can monitor system use and identify any irregularities for audit purposes.	Store Owner

8. Initial Product Backlog

Epic	Story ID	Summary	Points	Hours	Priority
EP-01	US-01	Admin: Add product (men's / boys' clothing / perfume) with images, description, price, stock	5	10	High
EP-01	US-02	Admin: Edit existing product details	3	6	High
EP-01	US-03	Admin: Delete product from catalogue	2	4	High
EP-01	US-04	Admin: Organise products into categories and apply tags	3	6	High
EP-01	US-05	Customer: Browse and filter products by category, price range, name	5	10	High
EP-01	US-06	Customer: View detailed product page with images and stock status	3	6	High
EP-01	US-07	System: Auto In Stock / Out of Stock status from inventory	2	4	High
EP-02	US-08	Customer: Add/update/remove products from shopping cart	5	10	High
EP-02	US-09	Customer: Generate pre-filled WhatsApp message for checkout	5	10	High
EP-02	US-10	System: Auto-create order record (Pending) on checkout initiation	3	6	High
EP-02	US-11	Admin: View all customer orders with statuses on dashboard	3	6	High

EP-02	US-12	Admin: Update order status (Pending → Confirmed → Processing → Ready for Delivery)	3	6	High
EP-02	US-13	Customer: View order history and current status up to dispatch	3	6	Medium
EP-03	US-14	Admin: Assign delivery person to a packed order	3	6	High
EP-03	US-15	Delivery Person: Accept order and update delivery status (Picked Up → Out for Delivery → Delivered)	5	10	High
EP-03	US-16	Customer: Track real-time delivery status (Assigned → Picked Up → Out for Delivery → Delivered)	3	6	High
EP-03	US-17	Customer: Submit star rating and review for a received product	3	6	Medium
EP-03	US-18	Customer: View approved reviews and aggregate ratings on product pages	2	4	Medium
EP-03	US-19	Admin: Approve, reject or delete reviews from moderation panel	3	6	Medium
EP-03	US-20	Admin: Manage static page content (About, Contact)	2	4	Low
EP-04	US-21	Admin: Create and manage system user accounts (CRUD)	3	6	High
EP-04	US-22	Admin: Assign roles and permissions to system users	3	6	High
EP-04	US-23	Admin: Unified dashboard with key operational metrics	3	6	Medium
EP-04	US-24	Admin: Configure store settings (name, contact, WhatsApp number)	2	4	Medium
EP-04	US-25	Admin: View system activity logs for audit purposes	2	4	Low
		TOTAL	79	158	

9. Definition of Done (DoD)

General Definition of Done (Project-Wide)

A user story or feature is considered Done when ALL of the following criteria have been met:

- All acceptance criteria defined for the user story have been implemented and verified through developer testing.
- The feature has been functionally tested by the developer and at least one other team member, with no critical or high-priority bugs outstanding.
- The code has been peer-reviewed and merged into the main development branch via a pull request on GitHub.
- The feature integrates correctly with all dependent modules and does not break existing functionality (regression testing passed).
- All relevant database schema changes have been applied and data integrity has been verified.
- The UI is responsive and functions correctly on both desktop and mobile browser viewports.
- API endpoints return correct HTTP status codes and responses, verified through Postman integration testing.
- Role-based access control is enforced: only authorised users can access or perform protected operations.
- The feature has been demonstrated to and accepted by the Scrum Master or Product Owner in a sprint review.
- Relevant documentation (inline code comments, API notes) has been updated to reflect the implementation.

Epic-Specific Definition of Done: EP-03 – Delivery Tracking with Review Management

In addition to the general DoD, EP-03 is considered Done when:

- An order assigned to a delivery person progresses correctly through all four delivery statuses (Assigned → Picked Up → Out for Delivery → Delivered), with each status change reflected immediately in the customer-visible order tracking view.
- A customer-submitted review is stored as Pending Moderation and only appears publicly on the product page after explicit admin approval, with the aggregate rating updating automatically upon publication.
- All moderation actions (approve, reject, delete) are recorded in the moderation activity log, as verified by database query validation.

10. Sprint 1 Preview

Sprint 1 is focused on establishing the core project infrastructure and delivering the foundational product and catalogue management functionality. The selected user stories represent all seven EP-01 stories (100% of the epic) and the access-control foundation from EP-04, providing a working base upon which all subsequent sprint work will build.

Story ID	Epic	User Story Summary	Priority	Key Acceptance Criteria
US-01	EP-01	Admin: Add product with images, description, price, stock	High	Product saved & visible on storefront; image stored in R2
US-02	EP-01	Admin: Edit existing product details	High	Updated details reflected immediately on storefront
US-03	EP-01	Admin: Delete product from catalogue	High	Product no longer visible to customers after deletion
US-04	EP-01	Admin: Organise products into categories and apply tags	High	Products filterable by category and tag on customer side
US-05	EP-01	Customer: Browse and filter by category, price range and name	High	Filtered results accurate; update without full page reload
US-06	EP-01	Customer: View detailed product page with images and stock status	High	Stock status correct; all images render from R2
US-07	EP-01	System: Auto In Stock / Out of Stock from inventory levels	High	Status updates automatically when stock quantity reaches zero
US-21	EP-04	Admin: Create and manage system user accounts	High	New account appears in user list; login succeeds with correct role
US-22	EP-04	Admin: Assign roles and access permissions to system users	High	Staff account restricted to permitted modules only

Sprint 1 also includes non-story sprint backlog items: repository initialisation, development environment configuration, database schema design (including delivery tracking and moderation tables), and Cloudflare R2 bucket configuration. Sprint 1 covers all 7 EP-01 stories (100%) and 2 of the 5 EP-04 stories (40%), with remaining EP-04 items provisionally planned for Sprint 4.

11. Software and Hardware Requirements

Software Requirements

Category	Tool / Technology	Purpose
Frontend Framework	React.js (v18+)	Customer storefront and admin dashboard UI development
Backend Framework	Node.js + Express.js	RESTful API development and business logic processing
Database	MySQL	Relational data storage for products, orders, delivery records, users and reviews
Object Storage	Cloudflare R2	Product image and media file storage and delivery
Authentication	JSON Web Tokens (JWT)	Stateless user authentication and role-based access control
Version Control	Git + GitHub	Source code management, branching and team collaboration
IDE / Code Editor	Visual Studio Code	Primary development environment for all team members
UI/UX Design	Figma	Wireframe and prototype design for customer and admin interfaces
API Testing	Postman	Backend API endpoint testing and validation
Diagram Tool	Draw.io / Lucidchart	UML diagrams, architecture diagrams and workflow charts
Package Manager	npm	Node.js dependency management

Hardware Requirements

Category	Specification	Purpose
Development Workstations (×3)	Intel Core i5 or equivalent, 8 GB RAM minimum, 256 GB SSD, 64-bit OS	Primary development machines for each team member
Mobile Device for Testing	Android or iOS smartphone with WhatsApp installed	Functional testing of WhatsApp checkout and delivery tracking workflows
Application Hosting (Cloud)	Cloud PaaS platform (e.g., Render, Railway, or Vercel)	Frontend and backend deployment to a live public URL
Database Hosting (Cloud)	Managed MySQL instance (e.g., Supabase or Neon free tier)	Hosted relational database service for production data
Internet Connection	Stable broadband connection (minimum 10 Mbps)	Development, testing and cloud service access

12. Budget

The following cost estimate covers the thirteen-week project duration. Free-tier services are used wherever possible; estimated commercial costs are noted for academic transparency. All figures are in Sri Lankan Rupees (LKR).

Item	Description	Est. Cost (LKR)	Notes
Domain Name	Custom .com or .lk domain for live deployment URL (1-year registration)	LKR 3,500	One-time cost
Cloudflare R2 Storage	Object storage for product images; 10 GB free tier sufficient for project scope	LKR 0 – 800/mo	Free tier likely
Application Hosting – Frontend	React.js frontend deployment via Vercel or Netlify free tier	LKR 0	Free tier
Application Hosting – Backend	Node.js/Express API deployment via Render or Railway starter tier	LKR 0 – 2,500/mo	Free tier sufficient

Database Hosting	Managed MySQL via Supabase or Neon free tier (up to 512 MB)	LKR 0	Free tier
Software Licenses	VS Code, Figma free tier, Postman, Draw.io, Git/GitHub all free or open-source	LKR 0	Open-source
Development Labour	3 team members × 13 weeks academic project, no commercial labour cost	LKR 0	Academic project
Contingency (10%)	Reserve for unexpected paid tier upgrades during development or testing	LKR 650	10% of hard costs
TOTAL (Estimated)		LKR ~7,450	

The primary hard cost is domain registration (approximately LKR 3,500 as a one-time expenditure). All software development tools are available at no cost, making this project highly feasible within the constraints of a student academic budget.

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14. Appendix

Appendix A – Glossary of Terms

Term	Definition
Gen-Z	The client organisation — a men's, boys' clothing and perfume retail store in Sri Lanka for which this platform is being developed.
Epic	A large body of work in Scrum methodology that can be broken down into smaller user stories.
User Story	A short description of a feature told from the perspective of the person who desires the capability, in the format: As a [role], I want [goal], so that [reason].
Product Backlog	An ordered list of everything needed in the product, managed by the Product Owner.
Sprint	A time-boxed development iteration in Scrum, typically 1–3 weeks in duration.
Definition of Done (DoD)	A shared understanding of what it means for a user story or feature to be complete and ready for delivery.
WhatsApp Checkout	The custom checkout mechanism where shopping cart contents are compiled into a pre-formatted WhatsApp message routed to the store owner for order confirmation.
Delivery Status Workflow	The sequence of delivery statuses tracked in EP-03: Assigned → Picked Up → Out for Delivery → Delivered.
RBAC	Role-Based Access Control — restricts system access to users based on their assigned role.
JWT	JSON Web Token — used for stateless user authentication and role-based access control.
CRUD	Create, Read, Update, Delete — the four basic operations performed on data in a database-backed application.
Cloudflare R2	A cloud-based S3-compatible object storage service used to store and serve product images.
RESTful API	Representational State Transfer API — an architectural style for networked applications using HTTP methods.

Appendix B – Project Team Signatures

We certify that this report does not contain any material previously published or written by another person except where due reference is made in the text.

Student ID	Name	Role	Signature
IT24101208	Warnakula S P	Scrum Master	_____
IT24101437	Sashorinth N	Team Member	_____
IT24103841	Razni Ahamed B S	Team Member	_____